

1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

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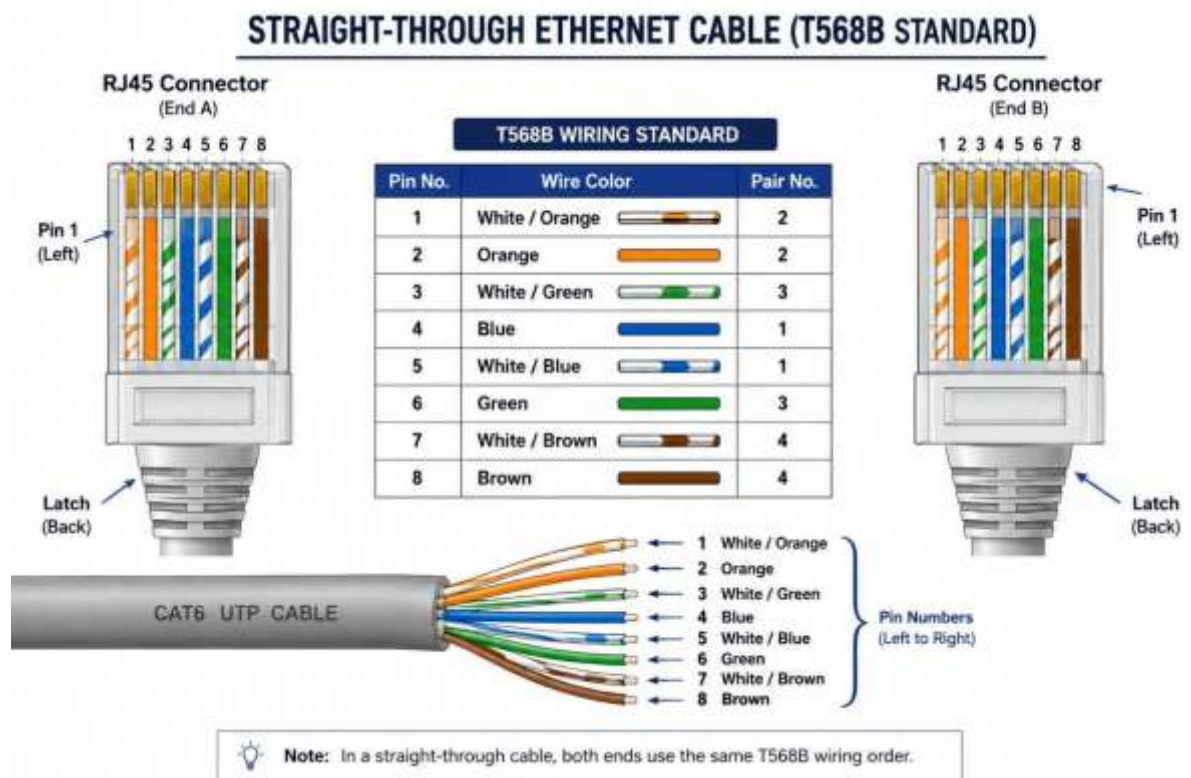
Feature	Ethernet Cable	Fiber Optic Cable
Transmission	Uses electrical signals through copper wires.	Uses light signals through glass/plastic fibers.
Speed & Distance	Good speed, but generally shorter distances.	Very high speed and works over much longer distances.
Cost	Cheaper and easier to install.	More expensive and requires specialized equipment.

Real-world examples:

- Ethernet: Connecting PCs to a switch in a gaming café or home router.
- Fiber optic: Connecting different buildings in a college campus or a data-center backbone.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

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3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

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Steps to Attach an RJ45 Connector to a CAT6 Cable

1. **Cut the cable:** Cut the CAT6 cable to the required length using a cable cutter.
2. **Remove the outer jacket:** Carefully strip about 2–3 cm of the outer insulation without damaging the internal wires.
3. **Separate the wires:** Untwist and straighten the four twisted pairs.
4. **Arrange the wires:** Follow the **T568B** color order:
White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
5. **Trim the wires:** Align all eight wires evenly and cut their ends straight.
6. **Insert into RJ45 connector:** Push the wires fully into the RJ45 plug, making sure each wire reaches its correct pin.

7. **Check the order:** Look through the transparent connector and confirm that the wires are in the correct order.
8. **Crimp the connector:** Insert the RJ45 plug into the **RJ45 crimper** and squeeze firmly to lock the contacts onto the wires.
9. **Test the cable:** Use a LAN cable tester to check that all eight connections are working correctly.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Hint: Consider speed, distance, and budget for your answer.

ANS:-

Gaming Cafe with 10 PCs

I would choose Ethernet CAT6 cables because 10 PCs in a gaming cafe usually do not require the long-distance capability of fiber optics. CAT6 provides sufficient speed and low latency for LAN gaming while being cheaper and easier to install. I would use an RJ45 crimper to attach the connectors because Ethernet cables use RJ45 connectors, whereas a fiber cleaver is used for preparing fiber-optic cables. This setup is more practical and budget-friendly for a small gaming café.